



Ramco Institute of Technology



TECH-EEE

Where Great minds share & glow

EMPOWER STUDENTS TO
ACHIEVE ACADAMIC
EXCELLENCE&INNOVATION



ACADEMIC YEAR
2020-2021

ABOUT

RAMCO INSTITUTE OF TECHNOLOGY



Ramco Institute of Technology is founded with a vision to impart high quality engineering education at an affordable cost. Under the able guidance of our Chairman Shri P. R. Ramasubrahmaneya Rajha, son of Shri P A C Ramaswamy Raja; distinguished professionals, academicians and education experts, we continually aim to revolutionize the learning environment by creating an enviable knowledge pool of engineering and technology graduates who are attuned to the current industry requirements.

Being part of the Ramco Group, which is well known for its qualitative and innovative brands not only in India but across the world, we are looked upon to set high standards in the education sector. True to its legacy, RIT has embarked on a mission to empower students with high-quality, accessible yet world-class engineering education and prepare these young minds for lifelong learning by creating and disseminating appropriate knowledge.

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

The Department of Electrical and Electronics Engineering is established in the year 2013. Currently, it offers a 4 year Under Graduate programme, B.E. Electrical and Electronics Engineering with a sanctioned intake of 60. The objective of the Department is to impart quality education and competitive training that will transform Students into Professional Engineers who are prepared to meet the needs of the society and adapt to the rapidly changing technology in various areas of Electrical and Electronics Engineering.

The Department has well-qualified and experienced Faculty members with specialization in Power Systems, High Voltage Engineering, Power Electronics and Control Systems. The Faculty members also have wide experience in teaching. They have a strong sense of responsibility to provide the finest possible engineering education to the students.

The Department has dedicated and state-of-the-art laboratories and leading softwares like DIgSILENT, NI-LabVIEW Academy, ETAP, MATLAB, PLC/SCADA and Power World Simulator to cater to academic and research activities. It provides the Students with an in-depth knowledge in fundamentals and practical applications, so that he/she can pursue for wide-range of career opportunities. The Department conducts Workshops, FDP, STTP, Value Added Courses, Seminars and Conferences and invites renowned academicians and industrial experts to enable the Students and Faculty to update with the latest developments. It also encourages Students to interact with experts from Industries and other prominent Institutions.

Vision, Mission, PEOs & PSOs

Vision

To produce highly competent Electrical and Electronics Engineers with professional skills and a sense of ethics to meet the demands of technologically advancing society.

Mission

- Empowering the students with up-to-date knowledge and technological skills in Electrical and Electronics Engineering thrust areas.
- Providing a rigorous practical training through industrial collaboration to produce Electrical and Electronics Engineers with good design experience.
- Adopting effective teaching learning methodologies and value added courses to create Electrical and Electronics Engineers with a strong fundamental knowledge.
- Enabling the students to be “life-long” learners to meet intellectual, ethical and career challenges.

Program Educational Objectives (PEOs)

- Graduates will have adequate knowledge in basic science, mathematics and engineering necessary to formulate, solve and analyse electrical and electronics problems.
- Graduates will have the ability to apply engineering & management skills in their own / team projects in multidisciplinary environment, leading to successful careers and research.
- Graduates will have ability to engage in lifelong learning and an appreciation for professional ethical behaviour.
- Graduates will demonstrate technical knowledge and ethical values for professional development to meet the societal needs.

Program Specific Outcomes (PSOs)

- Model and analyze the performance of Electrical Machines, Control and Instrumentation systems and Power and Power Electronic systems.
- Design the hardware and software requirements for the development of Electrical drives and Industrial Automation systems.
- Provide socially acceptable technical solutions to complex electrical engineering problems with the application of modern and appropriate techniques for sustainable development.
- Apply the knowledge of ethical standards and management principles required to work in a team.

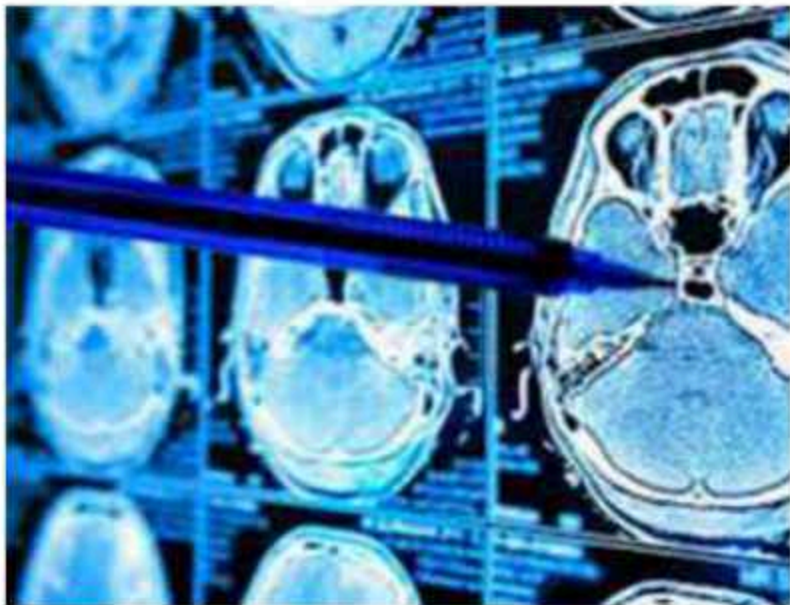


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ARTIFICIAL INTELLIGENCE IN MEDICINE

Artificial Intelligence in medicine may be characterized as the scientific discipline pertaining to research studies, projects, and the applications that aim at supporting decision based medical tasks through knowledge or data-intensive computer-based solutions that ultimately support and improves the performance of a human care



Artificial Intelligence (AI) research within medicine is growing rapidly. In 2016, health care AI projects attracted more investment than AI projects within any other sector of the global economy. However, among the excitement, there is equal skepticism, with some urging caution at inflated expectations. This article takes a close look at current trends in medical AI and the future possibilities for general practice

CURRENT TRENDS IN MEDICAL AI

New technologies entering the medical field must also integrate with current practices, gain appropriate regulatory approval, and perhaps most importantly, inspire medical staff and patients to invest in a new paradigm. These challenges have led to several emerging trends in AI research and adoption.

AI SUPPORTS POORLY RESOURCED SERVICES

A single AI system can support a large population and therefore it is ideally suited to situations where human expertise is a scarce resource.

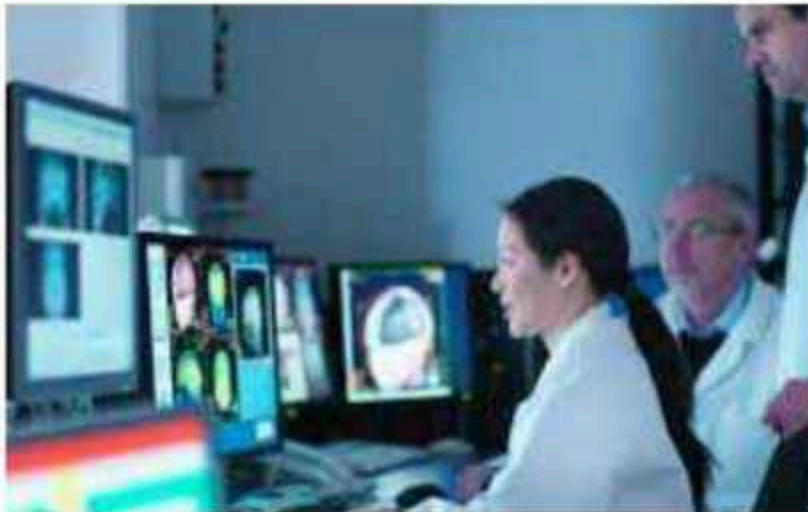


In many TBS prevalent countries there is a lack of radiological at remote centers could be interpreted by a single central system; a recent study shows that AI correctly diagnoses pulmonary TB with a sensitivity of 95% and specificity of 100%. Furthermore, under resourced tasks where patients are experiencing unsatisfactory waiting times are also attractive AI in the form of triage systems.

AI is a very picky eater

Developing ML models requires well-structured training data about a phenomenon that remains relatively stable over time. A departure from this result in 'over-fitting', where AI gives undue importance to spurious correlations within past data. In 2008, Google tried to predict the seasonal prevalence of influenza using only the search terms entered its search engine. Because peoples searching habits change dramatically with every passing year, the model was so poor predictive of the future that it was quickly discontinued

Prospects of AI in health care



The future of AI in health care could include tasks that range from simple to complex. Population health trending and analytics, therapeutic drug and device design reading, radiology images, making clinical diagnoses and treatment

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BY,

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II-YEAR,EEE.



BRAIN-COMPUTER INTERFACE (BCI)

By sunilram 2nd year EEE

Can you imagine that just by thinking you can do something. if you wanna type something you need not use your hands if you think it will happen.

Our brain consists of 86 billions of neurons. if you want to walk your brain sends electrical impulses to your legs and this electrical impulses helps to do what you think.

brain-computer interface (BCI) is a computer-based system that acquires brain signals, analyzes them, and translates them into commands that are relayed to an output device to carry out a desired action. In principle, any type of brain signal could be used to control a BCI system.

If you want to know about brain computer interface then you should know about EEG .

ELECTROENCEPHALOGRAPHY (EEG)
It is a physiological method of choice to record the electrical activity generated by the brain via electrodes placed on the scalp surface. In 1924, Berger was the first to record human brain activity by means of EEG. Berger was able to identify oscillatory activity, such as Berger's wave or the alpha wave (8–13 Hz), by analyzing EEG traces.

ELECTRODES
Electrodes are a crucial part of the brain-computer interface and the key to BCI technology. According to the electrode position and implantation method, the BCI can be divided into three ways, including non-intrusive electrodes, intrusive electrodes and semi-intrusive electrodes. electrodes are fixed in our head that helps to send the electrical impulses to the computer. These electrical impulses are decoded by computer and converted into message and displayed as output in screen.

INVASIVE BRAIN-COMPUTER INTERFACE

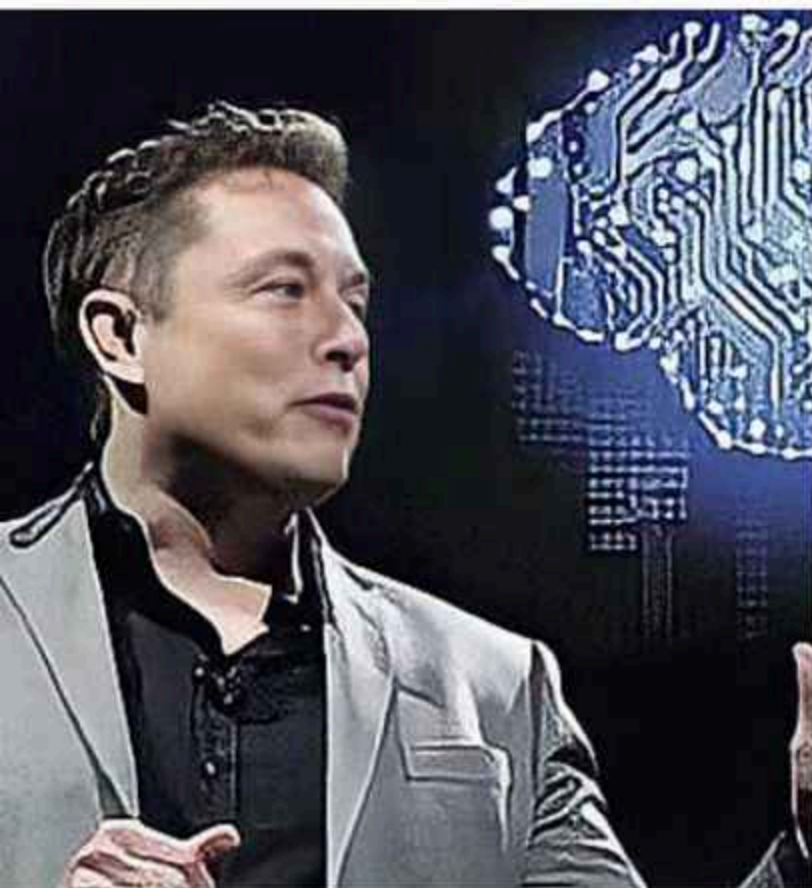
Invasive brain-computer interfaces (BCIs) have been developed to enable the direct communication between the brain and a computer or another external device. Unlike noninvasive BCI that have a lower spatial resolution, invasive BCI have the potential to record the activity of single neurons. In this surgery is required to implant electrodes.

NONINVASIVE BRAIN-COMPUTER INTERFACE

A noninvasive brain-computer interface (BCI) is a device that allows users to send messages or commands to devices, friends, family, or others through direct, noninvasive measures of brain activity.

Noninvasive BCIs record information from sensors placed on or very close to the head. No surgery is required to implant recording devices, and noninvasive BCIs do not use any painful or hazardous methods. The most common BCI recording mechanism is the electroencephalogram (EEG)

"A YEAR SPENT IN
ARTIFICIAL
INTELLIGENCE IS
ENOUGH TO
MAKE ONE
BELIEVE IN GOD."



ELECTRIC TRACTOR



Nowadays world can be slowly moved in E-vehicles to reduce global warming. As per a recent survey on a global scale, only 1 in 250 cars on the road is electric, Mean electric vehicles account for only 2.2% of the global vehicle market share. In India took only 1.3 percent of total vehicles sales in the country

Sonalika tractors were introducing the first tiger electric tractor in India. sonalika tractors have a manufacturing unit in Punjab, India. it is the 3rd largest manufacturer of India, with a capacity of producing 3lakh tractors per year. It has equipped with an Etrac motor that is claimed to offer high power density and high peak torque with zero RPM drop for optimal performance.

The new Tiger electric tractor can claim a top speed of 24.93kmph. Tiger electric tractor is powered by an energy-efficient etrac motor and offers 100% of torque availability at all times. Low cost running. Sonalika electric tractor loaded with an IP67(Lithium-ion battery) Complaint 25.5kw .

The compact battery offers a 75% lesser cost of running compared to diesel engine tractors. Tiger electric tractor can be charged with 3-pin 15-amp socket at home. It will take around 10 hours to fully charge. Company is also working on new charging system that would let the tractor charge in 4 hours. It's have excellent battery backup of 8 hours while operating with a 2-tonne trolley.

The tiger electric has been designed to offer seamless power with absolutely zero noise. The electric tractor going to introduce a concept of noiseless farming . Many states in India provide free electricity for farmers, so it will save the farmers investment.

ELECTRIC VEHICLES ARE NOT GOING TO TAKE THE MARKET BY STORM, BUT IT'S GOING TO BE A GRADUAL IMPROVEMENT.

By
jeyasuriya.S
2nd year EEE



EVOLUTION OF ELECTRIC VEHICLE

Where it all started? In Future,"EV are the greatest trends",



HISTORY BEHIND ELECTRIC VEHICLES:

But do we know about the History of Electric Vehicles? It starts from 18th century, In 1828 Anyos jedlik invented an electric motor and using this motor, he created the small electric car. Between 1832 and 1839, Scottish inventor Robert Anderson also developed a Crude electric carriage. In 1996, the First electric vehicle was a three-wheeler, invented by Scooter's India Private Limited and it was named as VIKRAM SAFA.

In 2000, BHEL developed an eighteen seater electric bus and then approximately 200 electric vans were made and run in Delhi. Electric vehicles came into Existence in the 19th Century, due to the high cost, low speed and short range. So, initially the demand declined worldwide.

Fast forward to 2021, The Largest source of Climate change is "Transportation". To solve the climate crisis, we need to make the vehicles on our roads as clean as possible. We have only one decade left to change the way we use energy to avoid the worst impacts of climate change.



GOVERNMENT INITIATIVE:

The Government of India has undertaken multiple initiatives to promote manufacturing and adoption of electric vehicles in India. With support of the Government, Electric vehicles have started penetrating the Indian market. Where companies are working aggressively to develop affordable electric vehicles, with a vision to dispel the notion that electric vehicles are expensive.

The Government in India aims to make the arrangements and incentives so that the Battery Electric Vehicles will encourage more people to contribute to 25% of all new vehicle registration by 2024 currently, EV contribute to only 0.29% of the overall Registrations in India. This will happen that Electric Vehicles should be promoted in the country while diesel must be stopped from usage in automobiles. The Indian Industry has also how been asked to work towards making India an EV hub by 2025.

IMPACTS OF DIESEL ENGINE VEHICLES:

Emissions from diesel and gasoline vehicles are not good for our planet; they're bad for our health. The analysis shows that electric vehicles perform better than gasoline vehicles, in terms of greenhouse gases emissions, depletion of non-renewable resources and emissions of atmospheric pollutants affected urban areas. Thus the World Switches to Electric Vehicles. The Electric Vehicle Industry in India is far behind with less than 1% of the total vehicle sales. Currently Indian Roads are dominated by conventional vehicles and have approximately 0.4 million electric cars only. The Indian EV industry has been on the back seat due to various challenges.



Electric Vehicle are not just the waves of future, they are saving lives today:

they are Ecofriendly.Today TESLA, MERCEDES BENZ, AUDI, HYUNDAI, MAHINDRA, CHEVROLET, BMW and Renault are Manufacturing ground breaking Electric Vehicles.

prepared by:
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INDIA'S ELECTRICITY GENERATION AND CONSUMPTION

GENERATION:

India is the huge populated nation and has an electricity coverage of 99.7%. India's total electricity generation is 1,547TWh. India generated the electricity from the utilities is 1,372TWh. The electricity generated from the non-utilities is 175TWh.

POWER GENERATION BY POWER PLANT:

The total installed capacity of power plant in India is 370.11GW. The share of fossil fuel is 79.8% and the renewable energy is 17.3%.

BY SECTORS:

The central sector has generated 93,477MW total of 25.2% power.

The state sector has generate 103,322MW total of 27.9% power.

The private sector has generate 173,308MW total of 46.8% power.

BY THERMAL:

The total power generated by the Thermal Power Plant is 2,30,600MW power total of 62.8%.

Coal Power Plant generate 1,98,525MW power total of 54.2%.

Lignite Power Plant generate 6,610MW power total of 1.7%.

Gas Power Plant generate 24,937MW power total of 6.9%.

BY HYDRO:

The total power generated by the Hydroelectric Power Plant is 45,699MW total of 12.4%.

BY NUCLEAR:

The total power generated by the Nuclear Power Plant is 6,780MW total of 1.9%.

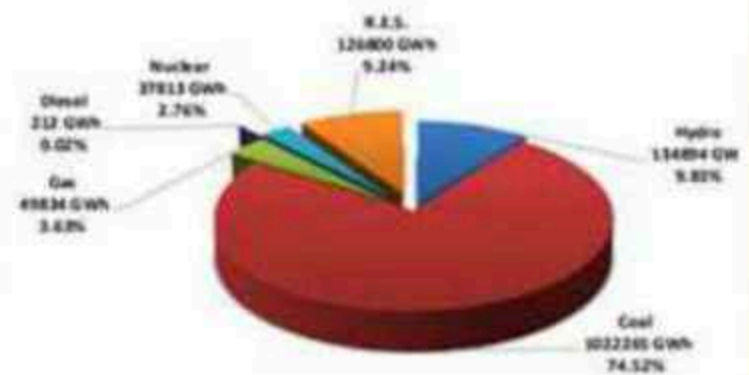
BY RENEWABLE ENERGY SOURCES:

Renewable Energy Sources include Small Hydro Project, Biomass Gasifier, Biomass Power, Urban & Industrial Waste Power, Solar and Wind Energy. The total power generated by this power plant is 87,028MW power total of 23.5%.

CONSUMPTION:

The India's gross electricity consumption is 1,181Kwh per capita in the year 2018-2019. The total power generated in the year 2018-2019 is 1371817Gwh.

The consumption of electricity in this year 2019-2020 is 1196309Gwh. The consumption data is only based on the 2018-2019, the no data available for the year 2019-2020. The no much more difference in consumption of electricity there is only increase of upto 10% in the year 2019-2020.

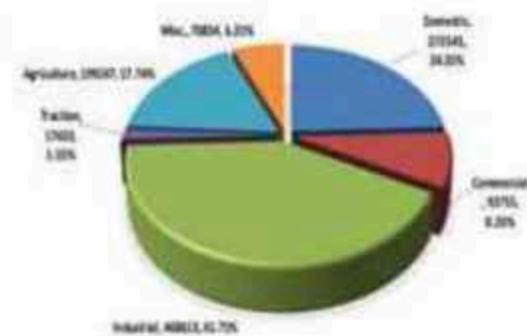
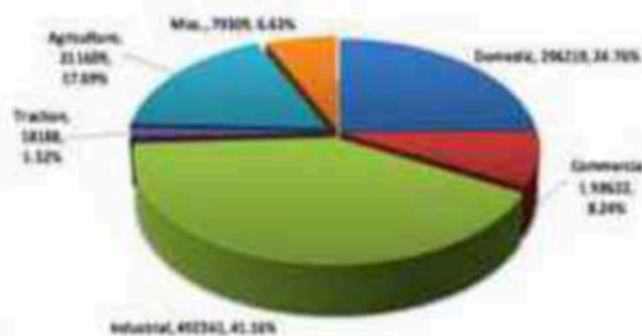


FORECAST DATA OF ELECTRICAL CONSUMPTION:

Category wise Forecast of Electrical Energy Consumption
For the Year 2019-20 and 2021-22
As Per 19th EPS (Volume-I)

Category	Year		GWh
	2019-20	2021-22	
Domestic	333356	386790	
Commercial	112630	128888	
Irrigation	247283	276277	
Industries	341492	386450	
Others	109818	122081	
Total	1144579	1300486	

ELECTRICAL CONSUMPTION:



CONCLUSION:

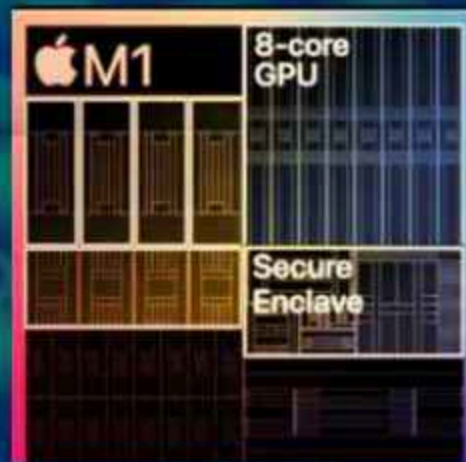
The Electricity demand is increasing day to day in Tamil Nadu. So, use the Electricity wisely and save the Electricity and save the future.

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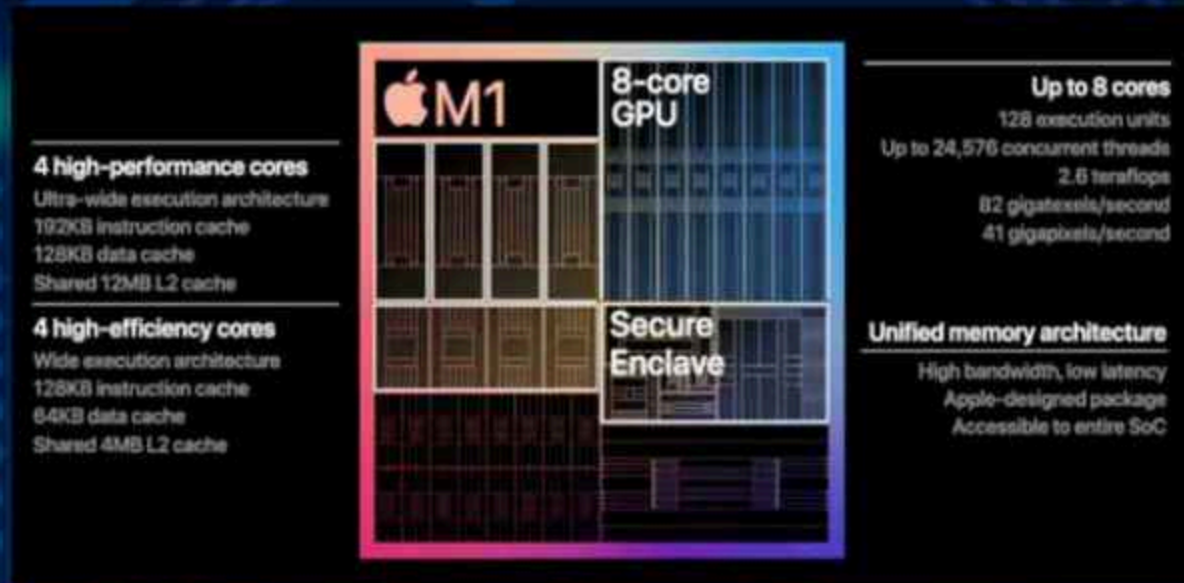
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<http://cea.nic.in/>

PRESENTED BY:
P.VISHNURAM
III- YEAR EEE

[REDACTED]



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[REDACTED]

ON THE HANDS OF THE DESIGNER
AND THE HANDS OF THE DESIGNER

[REDACTED]

Nano Diamond Battery

Modern life is heavily reliable on mobile battery-powered devices affecting daily aspects of our lives, ranging from telecommunication devices to transport vehicles. There is an increasing demand for efficient and cost-effective batteries. Conventional batteries have been riddled with numerous concerns



The Nano Diamond Battery (NDB) is a high-power, diamond-based alpha, beta, and neutron voltaic battery that can provide lifelong and green energy for numerous applications and overcome limitations of existing chemical batteries. The NDB acts like a tiny nuclear generator. The power source for the NDB is intermediate- and high-level radio isotopes that are shielded for safety by multiple levels of synthetic diamond. The energy is absorbed in the diamond through a process called inelastic scattering, which is used to generate electricity. The self-charging process will provide a charge for the full lifetime of any device or machine, with up to 28,000 years of battery life.

Since the battery is self-charging and requires only Exposure to natural air, any excess charge can be stored in capacitors, super capacitors, and secondary cells to extend battery life for cellphones, aircraft, rockets, electric vehicles, sensors, other devices and machinery.



Nuclear recycle process — Radioactive waste is reprocessed and recycled to enable sustainability and promote a clean energy source in a safe and secure environment.

Key innovations of NDB are sophisticated safety features covering thermal, mechanical, and radiation safety.

Lock-in system — Using a nuclear power source for a battery system brings up the question of nuclear proliferation due to production of fissionable isotopes such as Pu-238 and U-232. To tackle this issue, NDB uses an ion implantation mechanism, called a "lock-in system," that prevents usage other than power generation. This increases usability by meeting consumer safety requirements.



APPLICATIONS

Automotive — Electric vehicles have been heavily promoted by various governments and as such, it is one of the fastest-growing fields in recent years.

Industrial — NDB's safety, power output, and universality provide power to many routine applications and to those that are difficult to implement.

Aerospace — The aviation market is vast and many of the technological advancements come from the digital revolution.

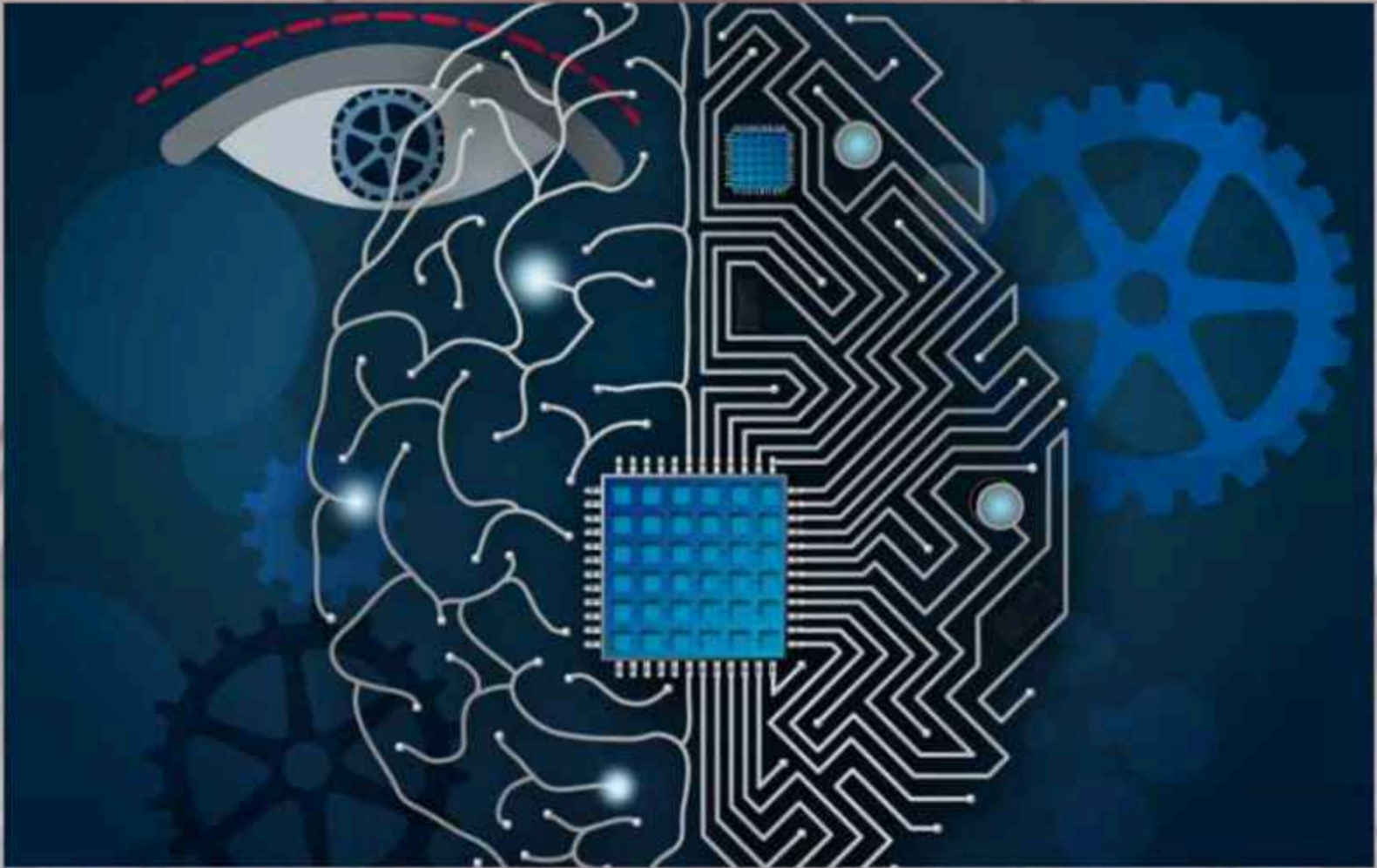
Medical Technology — In-situ medical devices and implantable such as hearing aids and pacemakers can benefit from long battery life in a smaller package with the added benefit of safety.

BY

P.Kartheeshwaran

II-YEAR EEE

NEURAL NETWORKS



What are neural networks?

Neural networks, also known as artificial neural networks (ANNs) or simulated neural networks (SNNs), are a subset of machine learning and are at the heart of deep learning algorithms. Their name and structure are inspired by the human brain, mimicking the way that biological neurons signal to one another.

Artificial neural networks (ANNs) are comprised of a node layers, containing an input layer, one or more hidden layers, and an output layer. Each node, or artificial neuron, connects to another and has an associated weight and threshold. If the output of any individual node is above the specified threshold value, that node is activated, sending data to the next layer of the network. Otherwise, no data is passed along to the next layer of the network.

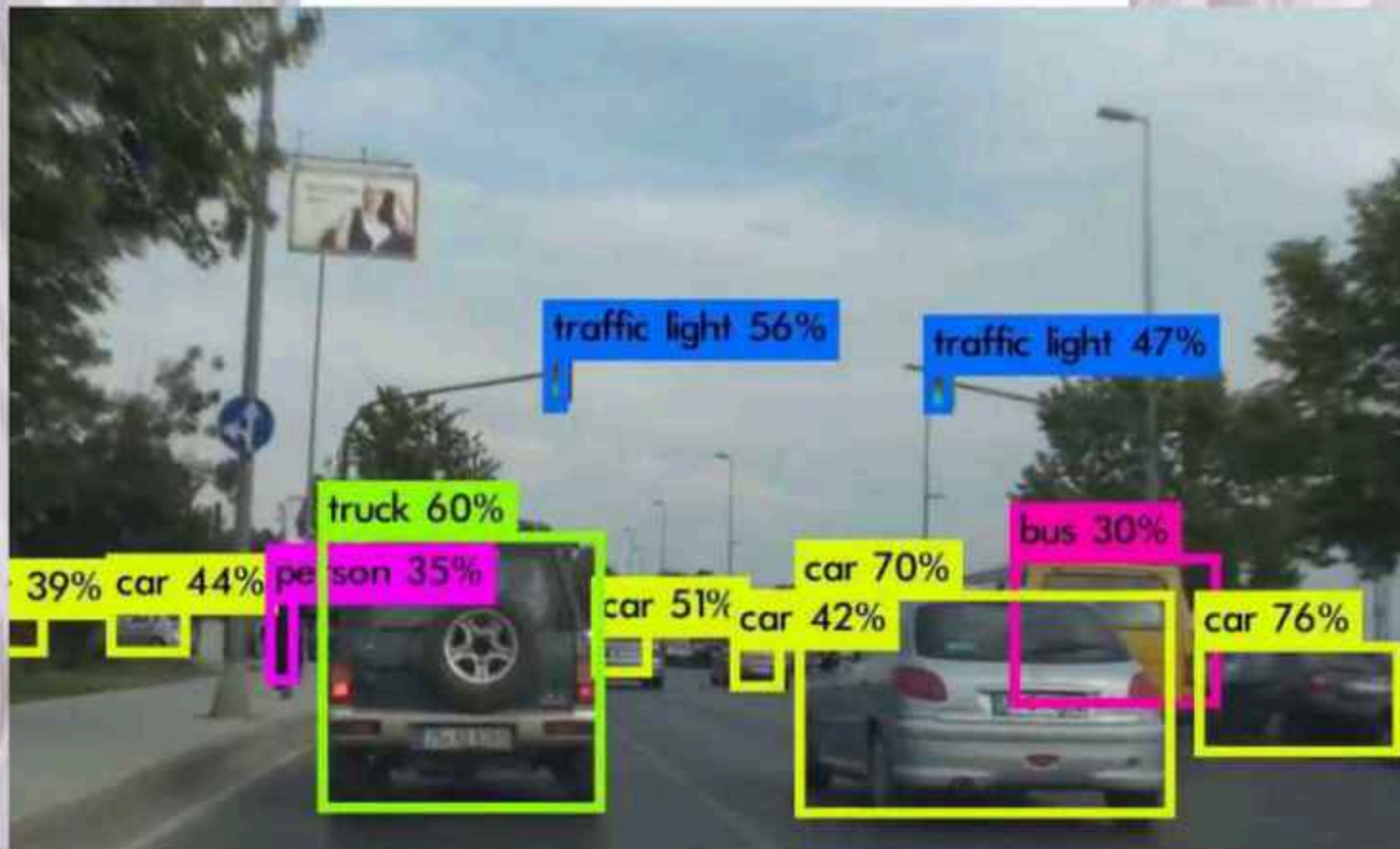
Neural networks rely on training data to learn and improve their accuracy over time. However, once these learning algorithms are fine-tuned for accuracy, they are powerful tools in computer science and artificial intelligence, allowing us to classify and cluster data at a high velocity. Tasks in speech recognition or image recognition can take minutes versus hours when compared to the manual identification by human experts. One of the most well-known neural networks is Google's search algorithm.

Neural networks reflect the behavior of the human brain, allowing computer programs to recognize patterns and solve common problems in the fields of AI, machine learning, and deep learning.

How do neural networks work?

Think of each individual node as its own linear regression model, composed of input data, weights, a bias (or threshold), and an output. The formula would look something like this:

$$\sum_{i=1}^n w_i x_i + \text{bias} = w_1 x_1 + w_2 x_2 + w_3 x_3 + \text{bias} \quad \text{output} = f(x) = \begin{cases} 1 & \text{if } \sum w_i x_i + b \geq 0 \\ 0 & \text{if } \sum w_i x_i + b < 0 \end{cases}$$



How does AI and Neural networks works in tesla,inc?

We develop and deploy autonomy at scale in vehicles, robots and more. We believe that an approach based on advanced AI for vision and planning, supported by efficient use of inference hardware, is the only way to achieve a general solution for full self-driving and beyond.

Apply cutting-edge research to train deep neural networks on problems ranging from perception to control. Our per-camera networks analyze raw images to perform semantic segmentation, object detection and monocular depth estimation. Our birds-eye-view networks take video from all cameras to output the road layout, static infrastructure and 3D objects directly in the top-down view. Our networks learn from the most complicated and diverse scenarios in the world, iteratively sourced from our fleet of nearly 1M vehicles in real time. A full build of Autopilot neural networks involves 48 networks that take 70,000 GPU hours to train . Together, they output 1,000 distinct tensors (predictions) at each timestep.

**'AI WILL BE THE BEST OR WORST THING EVER FOR HUMANITY,'
- ELON MUSK**

Keerthian
1st-year EEE

PEGASUS

PEGASUS IS SPYWARE DEVELOPED BY THE ISRAELI CYBERARMS FIRM NSO GROUP THAT CAN BE COVERTLY INSTALLED ON MOBILE PHONES (AND OTHER DEVICES) RUNNING MOST VERSIONS OF IOS AND ANDROID. THE 2021 PROJECT PEGASUS REVELATIONS SUGGEST THAT THE CURRENT PEGASUS SOFTWARE CAN EXPLOIT ALL RECENT IOS VERSIONS UP TO IOS 14.6. AS OF 2016, PEGASUS WAS CAPABLE OF READING TEXT MESSAGES, TRACKING CALLS, COLLECTING PASSWORDS, LOCATION TRACKING, ACCESSING THE TARGET DEVICE'S MICROPHONE AND CAMERA, AND HARVESTING INFORMATION FROM APPS. THE SPYWARE IS NAMED AFTER PEGASUS, THE WINGED HORSE OF GREEK MYTHOLOGY. IT IS A TROJAN HORSE COMPUTER VIRUS THAT CAN BE SENT "FLYING THROUGH THE AIR" TO INFECT CELL PHONES.



5 THINGS ABOUT PEGASUS:

1. PEGASUS IS ZERO DAY SPYWARE FOR IOS.
2. PEGASUS IS A LOW PROBABILITY, BUT HIGH THREAT IMPACT.
3. APPLE'S IOS 9.3.5 UPDATE WILL NOT DETECT OR REMOVE PEGASUS.
4. PEGASUS EXPOSES ALL MESSAGES, CALLS, EMAILS, DATA, COMMUNICATION, AUDIO, VIDEO ETC..
5. EXISTENCE OF OTHER EXPLOITS LIKE PEGASUS IS LIKELY.



STORY :

THE PLAYERS:	NSO Group Cyber war software	UAE (suspected) Nation state	Ahmed Mansoor Human rights activist	Citizen Lab Research laboratory	Apple Mobile devices
WHAT HAPPENED:	Found vulnerabilities in iOS (data's report) Pegasus, a next-day "useful intercept" spyware product for governments, exploits 3 iOS vulnerabilities to jailbreak and take over mobile devices.	Purchased Pegasus from NSO to spy on Ahmed Mansoor Sent an SMS message with a malicious link capable of completely compromising his mobile device.	Shortly, did not click on the SMS link Contacted Citizen Lab for forensic analysis	Recognized exploit as an NSO product Analyzed the exploit Contributed lookout for support in the analysis Notified Apple of vulnerabilities	Patched the three vulnerabilities and released iOS 9.3.5 update Filed CVE reports

USE OF SPYWARE :

- BAHRAIN
- INDIA
- SAUDI ARABIA
- MOROCCO
- UNITED ARAB EMIRATES
- MEXICAN DRUG CARTELS

REACTION OF MEDIA :

NEWS OF THE SPYWARE RECEIVED SIGNIFICANT MEDIA ATTENTION, PARTICULARLY FOR BEING CALLED THE "MOST SOPHISTICATED" SMARTPHONE ATTACK EVER, AND FOR BEING THE FIRST DETECTION OF A REMOTE APPLE JAILBREAK EXPLOIT.

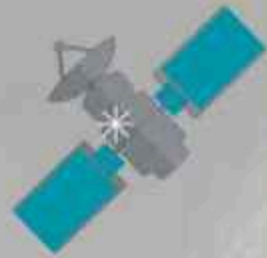
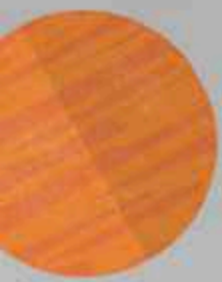
CONCLUSION :

THUS, THE PEGASUS PLAYS A VITAL ROLE IN HACKING.

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SENSORS:

A Sensor is a device that detects the change in the environment and responds to some output on the other system. A sensor converts a physical phenomenon into a measurable analog voltage converted into a human-readable display or transmitted for reading or further processing.

WORKING:

Sensor is a device or a module that helps to detect any changes in physical quantity like pressure, force or electrical quantity like current or any form of energy. After observing the changes, sensor sends the detected input to a microcontroller or microprocessor. Finally, a sensor produces a readable output signal, which can be either optical, electrical, or any form of signal that corresponds to change in input signal. In any measurement system, sensors play a major role. In fact, sensors are the first element in the block diagram of measurement system, which comes in direct contact with the variable to produce a valid output.

CLASSIFICATION OF SENSORS:

- Active and Passive Sensors
- Analog and Digital Sensors

“SOMETHING. OF COURSE! HE THOUGHT. HE HAD TOUCHED THE SCREEN. IT WAS A TOUCH SCREEN! THE RED LIGHTS AROUND THE EDGES MUST BE INFRARED SENSORS. TIM HAD NEVER SEEN SUCH A SCREEN, BUT HE'D READ ABOUT THEM IN MAGAZINES.”

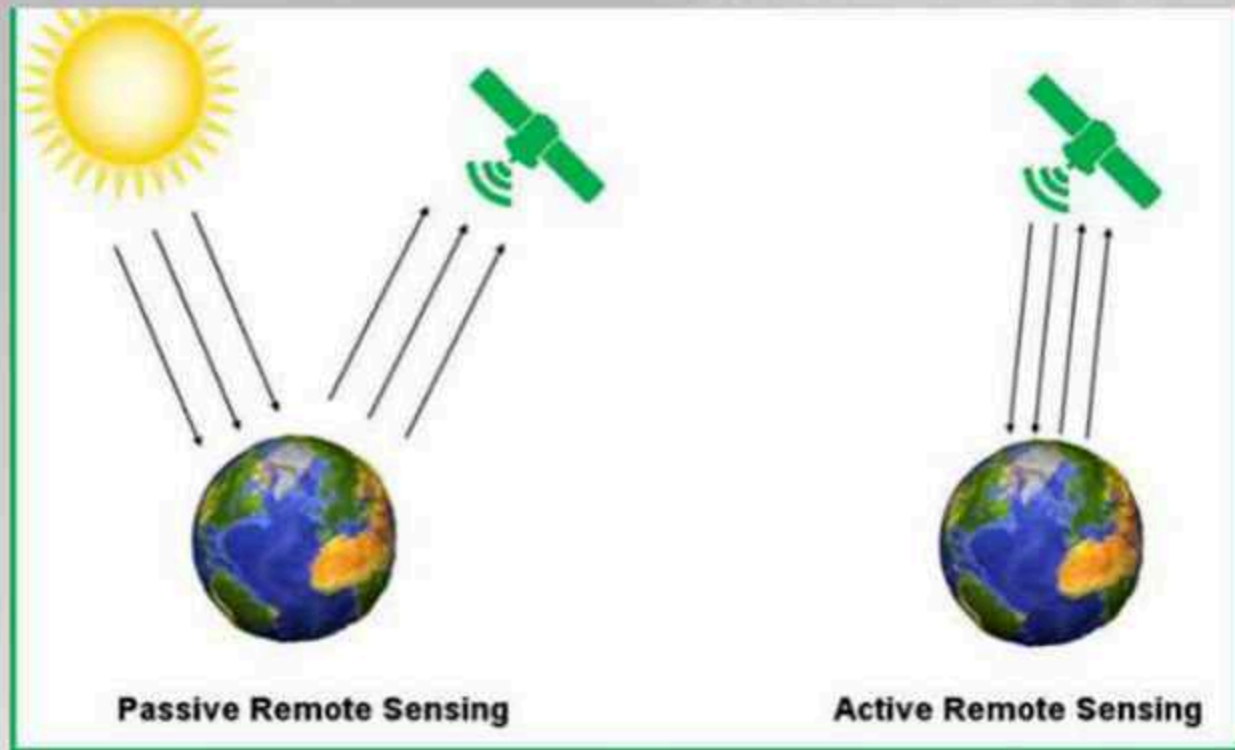


LESS THAN 1% OF \$50 MILLION ONLY 15%

ACTIVE SENSORS:

Active sensors are the type of sensors that produce output signals with the help of external excitation supply. The physical properties of the sensor varies with respect to applied external effect. Therefore, it is also called as Self Generating Sensors.

Example: LVDT and Strain gauge



PASSIVE SENSORS:

Passive sensors are the type of sensors that produce output signals without the help of external excitation supply.

Example: Thermocouple

ANALOG SENSORS:

The Sensor that produces continuous signal with respect to time with analog output is called Analog sensors. Generally, analog voltage ranges from 0-5 V.

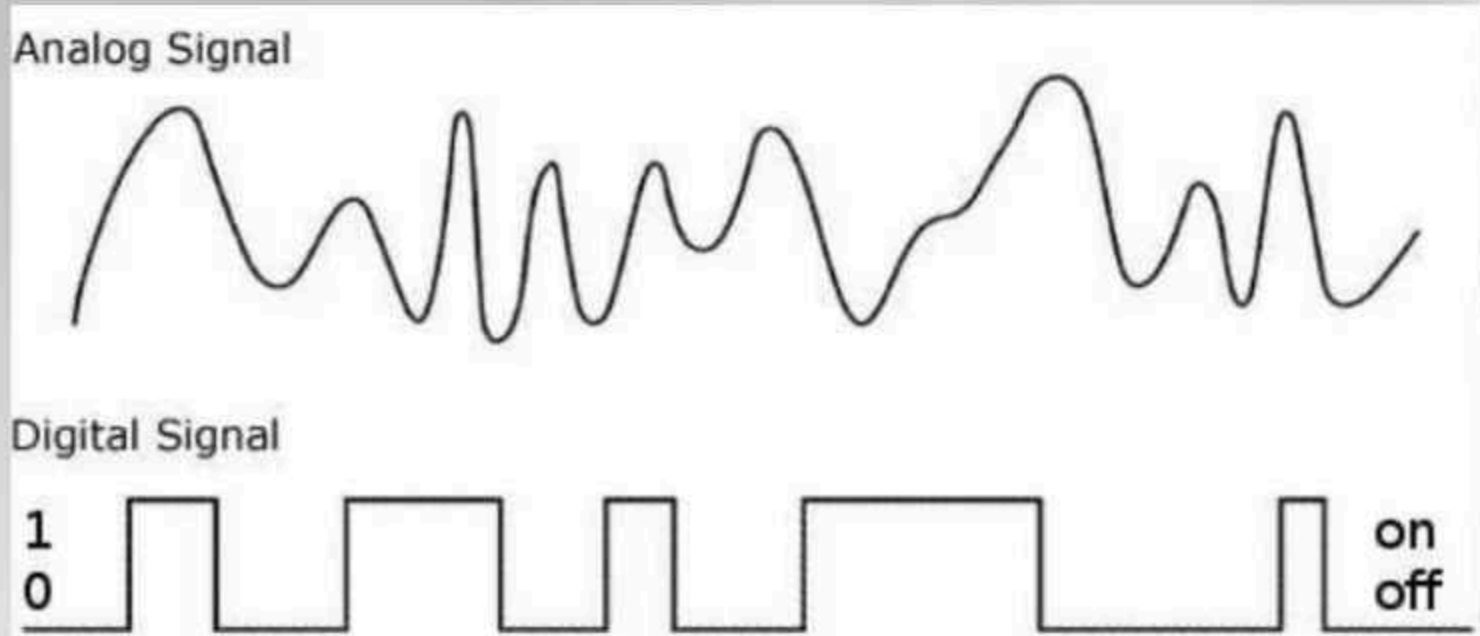
Example: Accelerometers, Speed sensor



DIGITAL SENSORS:



When data is converted and transmitted digitally, it is called Digital sensors. Digital sensors are the one which produces discrete output signals. Discrete signals will be non-continuous with time and it can be represented in bits for serial transmission and in bytes for parallel transmission. Digital output can be in form of logic 1 or logic 0 (ON or OFF).
Example: Light sensor



TYPES OF SENSORS:

- Thermocouples, RTDs and Thermistors: for measuring temperature.
- Strain gages: to measure strain on an object.
- Load cell: for measuring weight and load.
- LVDT Sensors: LVDTs are used to measure displacement in distance.
- Accelerometers: measuring vibration and shock.
- Microphones: for capturing sound waves.
- Current transducer: for measuring AC and DC current.
- Optical sensors: used to detect light, transmit data and replace conventional sensors.
- Camera sensors: used to capture single and continuous 2D images.



prepared by:

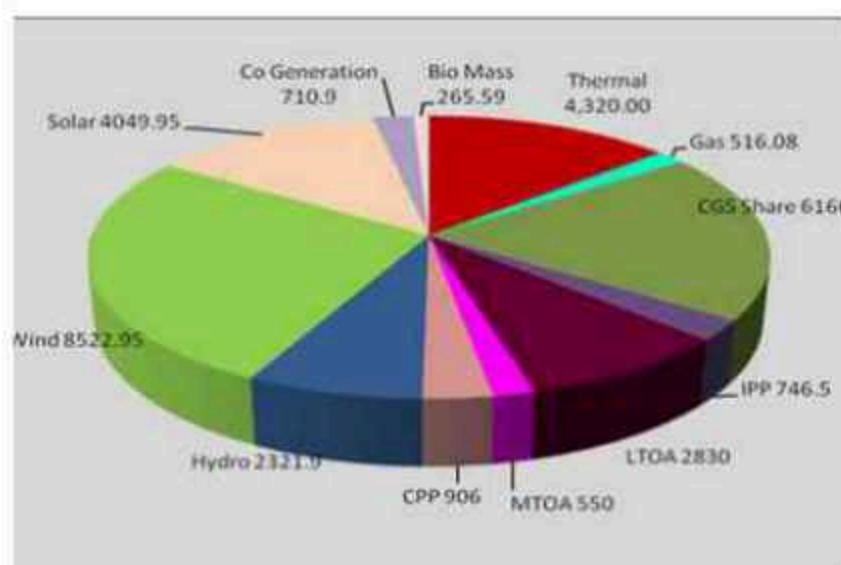
P.G.Gayathree , 1stYear EEE
G.Dhana Bharathi Lakshmi, 1stYear EEE

TAMILNADU ELECTRICITY GENERATION, DISTRIBUTION AND CONSUMPTION

- VISHNURAM 3RD YEAR EEE

Tamil Nadu Electricity Board (TNEB) was formed on July 1, 1957 under section 54 of the Electricity (Supply) Act 1948 in the State of Tamil Nadu as a vertically integrated utility responsible for power generation, transmission and distribution. The electricity network has since been extended to all villages and towns throughout the State. As per the provisions under the section 131 of the Electricity Act, 2003 TNEB was restructured on 1.11.2010 into TNEB Limited; Tamil Nadu Generation and Distribution Corporation Limited (TANGEDCO); and Tamil Nadu Transmission Corporation Limited (TANTRANSCO).

Peak Demand in MW



Transmission

In the Transmission process was handled by the Tamil Nadu Transmission Corporation Limited (TANTRANSCO). The vision of the TANTRANSCO was "To provide adequate and reliable transmission infrastructure at competitive rates". The all the power can be transmitted in the "State Load Dispatch Centre" (SLDC).

TAMILNADU ELECTRICITY GENERATION, DISTRIBUTION AND CONSUMPTION

Details	2010-11	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20
Installed capacity (MW) (Conventional Energy)	10,236.94	10,364.54	10,515.34	11,884.44	13,231.44	17,976.48	18,732.78	18,747.28	18,299.66	# 16,034.58
Installed capacity (MW) (Renewable Energy)	6,646.07	7,792.03	7,999.03	8,219.67	8,470.16	9,629.52	10,465.14	11,112.71	12,179.65	15,871.29
Total Installed capacity	16,883.01	18,156.57	18,514.37	20,104.11	21,701.60	27,606.00	29,212.39	29,859.99	30,479.31	31,905.87
Generation Purchase(MU)	&76,071	76,535	74,872	89,793	94,128	99,691	1,05,728	1,06,827	1,13,790	1,14,229
Energy Sales (MU)	58,858	61,387	58,810	70,976	76,828	79,464	83,020	85,890	91,918	94,944
Substations (Nos.)	1,343	1,320	1,341	1,392	1,442	1,499	1,547	1,592	1,682	1,784
EHT Lines (Lakh Ckt.Kms.)	1.77	1.80	1.84	1.89	1.95	2.00	2.04	2.09	0.34	0.35
HT lines (Lakh.Kms.)									1.77	1.81
LT lines (Lakh.Kms.)	5.56	5.67	5.78	5.88	5.98	6.10	6.19	6.28	6.28	6.35
Distn.Transformers (Nos.)	2,03,794	2,12,919	2,21,288	2,31,224	2,46,748	2,64,029	2,82,028	2,93,301	3,09,468	3,27,163
Per capita consumption (Units)	1040	1065	1011	1196	1228	1,280	1,340	1,389	1,467	1515
Towns, Villages Hamlets (Nos.)	&63,956	63,956	79,263*	79,248*	79,245*	79,245*	79,279*	79,283*	79,317*	79,360*
Pumpsets (Lakhs)	19.85	20.16	20.33	20.47	20.46	20.60	20.73	21.03	21.17	21.22
Consumers (Lakhs)	223.44	231.80	243.51	252.32	264.67	270.26	279.27	288.09	295.98	307.54

Functions of SLDC

Grid Operation in TN started by November 1964. The first Load Despatch Centre was operated from Erode. Subsequently, the main Load Despatch Centre was formed in 1986 at Chennai and Sub Load Despatch Centre at Madurai.

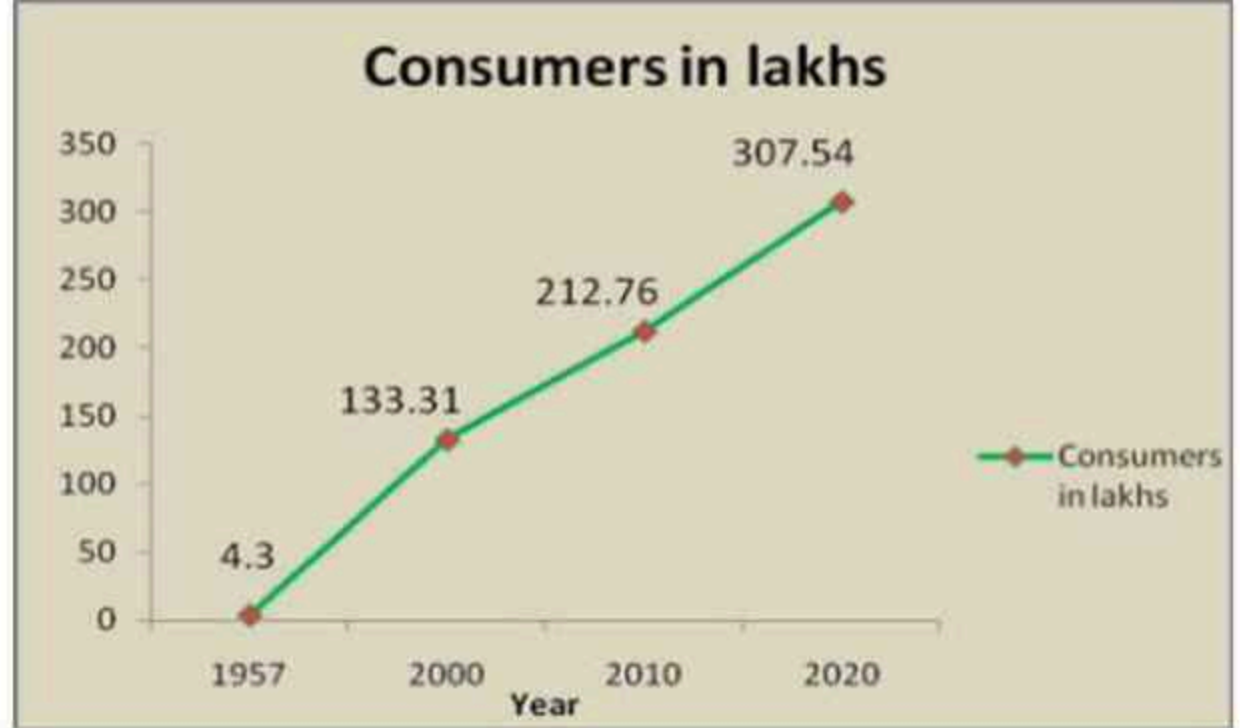
In accordance with section 32 of Electricity Act, 2003 roles and functions of SLDCs are as under:

- The SLDCs shall be the Apex Body to ensure integrated operation of the Power system in a State.
- SLDCs shall : Be responsible for optimum scheduling and dispatch of electricity within a state in accordance with the contracts entered into with the licensees or the generation companies operating in that State.
- Monitor grid operation Keep accounts of the quantity of electricity transmitted through State Grid.

Exercise supervision and control over the inter-state transmission system. Be responsible for carrying out real time operation for grid control and dispatch of electricity within the State through secure and economic operation of the State Grid in accordance with the grid standards and state grid code.

Distribution

The distribution process will be carried out by Tamil Nadu Generation and Distribution Corporation Limited(TANGEDCO). The total number of consumers being served in the State as on 31.03.2020 is 307.54 Lakhs. An efficient distribution network is needed for effectively utilizing the energy that is generated. TANGEDCO has an efficient distribution network that has grown over the years. The Consumption of the electricity was increased by the year and year in Tamil Nadu. In this generation all the peoples are used the electricity and the demand for electricity was keep on increasing.



Consumers	Tariff	Consumers as on 31.03.20 (in lakhs)	Consumption in MU (Provisional)
HT (Including sale to traders)	I to V	0.10	32,421
LT			
Domestic	I A	221.09	28,650
Huts	I B	10.09	490
Bulk Supply	I C	0.01	9
Public Lighting & Water supply	II A	7.46	2,304
Recognised Educational Institutions & Private Educational Institutions	II B (1) & II B (2)	0.95	454
Actual places of public worship	II C	1.51	141
Cottage and Tiny Industries	III A (1)	2.25	302
Power Looms	III A(2)	1.61	1,057
Industries	III B	3.25	7,314
Agriculture	IV	21.22	13,811
Commercial	V	34.30	7,725
Temporary Supply	VI	3.71	266
Total (HT & LT)		307.54	94,944

In Tamil Nadu the electricity generation is higher than the other state in India. The Electricity demand is increasing day to day in Tamil Nadu. So, use the Electricity wisely and save the Electricity and save the future



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